Step 1 — Import Libraries

```
import pandas as pd
import numpy as np

from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score
```

Step 2 — Load Dataset

```
data = pd.read_csv("flight.csv")
data = data.sample(5000)
```

Step 3 — Data Cleaning

```
data = data.dropna()
```

Step 4 — Encode Categorical Variables

```
data = pd.get_dummies(data)
```

Step 5 — Define Features and Target

```
# Create target column
data["delay"] = (data["arr_del15"] > 0).astype(int)

# Features and target
X = data.drop(["arr_del15", "delay"], axis=1)
y = data["delay"]
```

Step 6 — Normalize Data

```
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

Step 7 — Apply PCA

```
pca = PCA(n_components=10)
X_pca = pca.fit_transform(X_scaled)
```

Step 8 — Train Test Split

```
X_train, X_test, y_train, y_test = train_test_split(
    X_pca, y, test_size=0.2, random_state=42
)
```

Step 9 — Train Model

```
model = RandomForestClassifier()
model.fit(X_train, y_train)
```

```
▾ RandomForestClassifier
RandomForestClassifier()
```

Step 10 — Accuracy

```
y_pred = model.predict(X_test)

accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy)
```

```
Accuracy: 0.9638916750250752
```